

A FIELD LOG FOR THE NIGHT SKY

The Observatory

■ ■ ■



ORION · GATE OF THE WINTER SKY

Twelve month skies point to twenty accurate target cards – nebulae, clusters, galaxies, the planets, and the Moon – and every sight feeds a sketch-circle observation sheet and one growing life list.

Open the dome »

MONTHS · TARGETS · SESSIONS · LIFE LIST

First Light

The Observatory is written for observers at northern mid-latitudes – roughly 35 to 55 degrees north, the sky over most of North America, Europe, and Japan. Every month page lists deep-sky sights genuinely well placed that month; every card in the catalog is accurate and evergreen.

HOW TO WORK THE SKY

1. Open this month's sky page and read its highlight rows.
2. Tap any row – the target's card holds type, magnitude, and finder notes.
3. Star-hop with the card at the scope, then log the night on an observation sheet.
4. Sketch what the eyepiece actually shows in the circle – date it, note the sky.
5. Tick the life list as targets fall; each card's first-observed box remembers when.

WHERE TO BEGIN

[The worked example »](#)[Your observatory »](#)[January's sky »](#)[The life list »](#)

THE WANDERERS

The Moon and planets drift along the ecliptic on their own calendars, so the month pages name only the fixed deep sky. Check a current almanac or app for where the wanderers stand tonight – each has a card in the catalog.

THE ONE RULE

Never look at the Sun through any unfiltered instrument – not for a moment. The eclipse-safety card in the catalog is required reading before any solar attempt.

NOTES



The Worked Example

One night taken end to end – an observation sheet from a moonless January evening on the Orion Nebula, and the target card it filled in: first-observed box dated, notes written while the eyepiece was still frosting.

Everything on this bench is marked EXAMPLE. Your live log runs from Your Observatory.

ON THE BENCH

The worked session »

M42, card filled »

THE DOME

HOW THE WIRING WORKS

Month highlights are reference chips – each opens the same card the life list ticks, so a target's page is one tap from anywhere. Cards link back to the life list, observation sheets chain night into night, and every page returns here.

OBSERVING NOTES

A Night on the Orion Nebula

DATE

Sat 17 Jan 2026 · 21:30

SKY · CONDITIONS

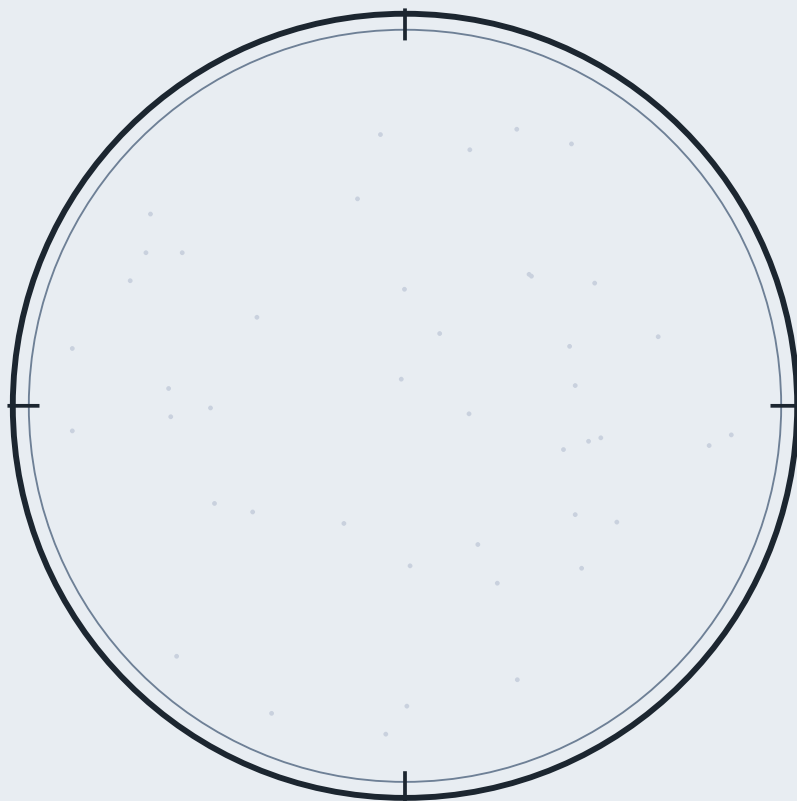
Clear and transparent – no Moon
all evening, hard frost

SEEING

Steady · stars calm at 100x

TARGET

M42 · Orion Nebula

[The catalog »](#)

SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

Started at 48x: the nebula's wings stretched over half the field with averted vision, a gray-green fan opening east. The Trapezium split clean at 96x – four stars in a tight keystone. Sketched the dark bay that cuts into the glow beside them. Best view of the winter so far; gloves off thirty seconds at a time, eyepiece frosting by eleven.



M42

Orion Nebula, Logged

Life list »

The winter showpiece, due south mid-evening

TYPE	Emission nebula
CONSTELLATION	Orion
MAGNITUDE	4.0 – visible to the naked eye
DIFFICULTY	Easy – rewards every instrument

FINDER NOTES

Below Orion's three-star Belt hangs the Sword. Its middle "star" looks fuzzy to the naked eye – that fuzz is the nebula. Any telescope shows gray-green wings of gas folded around the Trapezium, a tight quadrilateral of newborn stars at the heart. Start at low power for the full sweep, then magnify the Trapezium; averted vision doubles how far the wings reach.

FIRST OBSERVED

DATE

17 Jan 2026 · 21:30

INSTRUMENT · EYEPIECE

150 mm reflector · 25 mm and 12.5 mm eyepieces

SKY · SEEING

Clear, dark, moonless · steady seeing

NOTES

The wings ran farther than expected with averted vision. All four Trapezium stars held at 96x. The dark bay – the "Fish's Mouth" – cuts hard into the brightest glow. Came back inside grinning; this card is why the log exists.



Your Observatory

Twelve month skies over a twenty-target catalog – nebulae, clusters, galaxies, doubles, the planets, the Moon, and the eclipse-safety brief – with a life list to tick, 20 sketch-circle observation sheets, the equipment shelf, and a two-page glossary, all pre-linked.

This copy: 20 observation sheets. Set sessionCount (8-40) in the generator config.

ON THE BENCH

THE DOME

The twelve month skies »

The target catalog »

The life list »

Observation sheets »

The equipment shelf »

The glossary »

HOW THE WIRING WORKS

Month highlights are reference chips – each opens the same card the life list ticks, so a target's page is one tap from anywhere. Cards link back to the life list, observation sheets chain night into night, and every page returns here.

OBSERVING NOTES

January

Crisp, early dark, Orion due south by mid-evening. The Quadrantid meteors peak in the first days.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M42 · Orion Nebula, due south☐**M45 · Pleiades, near overhead**☐**The Double Cluster, high northwest**☐**M44 · Beehive, well up by nine**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[February »](#)

February

Orion still rules the south while the Beehive climbs the east and the first galaxies wait in the wings.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M42 · Orion Nebula, early evening☐**M45 · Pleiades, high southwest**☐**M44 · Beehive, riding high**☐**M81 · Bode's Galaxy, climbing northeast**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

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MOON & PLANET NOTES, THIS YEAR

[« January](#)[March »](#)

March

Equinox month – winter tips into the west while the spring galaxies rise behind it.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M44 · Beehive, due south mid-evening☐**M81 · with M82 in one field**☐**M51 · Whirlpool, up by ten**☐**M42 · last call, sinking southwest**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[« February](#)[April »](#)

April

Galaxy season proper – dark evenings between the Dipper and Virgo. Lyrid meteors late in the month.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M51 · Whirlpool, under the handle☐**M81 · Bode's Galaxy, near the zenith**☐**M104 · Sombrero, crossing south**☐**M44 · Beehive, high in the west**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[« March](#)[May »](#)

May

The Dipper hangs overhead, galaxies everywhere – and the first globular of summer rises in the east.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M13 · Hercules Cluster, rising east☐**M51 · Whirlpool, best placed**☐**M104 · Sombrero, early evening south**☐**M81 · Bode's Galaxy, still high**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

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MOON & PLANET NOTES, THIS YEAR

[<< April](#)[June >>](#)

June

Shortest nights – wait out the long twilight, then the summer sky arrives all at once.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M13 · Hercules Cluster, high east☐**M57 · Ring Nebula, under Vega**☐**Albireo · the gold and blue double**☐**M51 · Whirlpool, overhead, dark sky**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[<< May](#)[July >>](#)

July

The Milky Way stands up out of Sagittarius – nebula country, low and rich in the south.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET	EYEPIECE · DATE	SEEN
M8 · Lagoon, over the Teapot		☐
M20 · Trifid, beside the Lagoon		☐
M13 · Hercules Cluster, at the zenith		☐
M57 · Ring Nebula, high east		☐
M27 · Dumbbell, well up by eleven		☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

« June

August »



August

Perseid meteors peak mid-month and the Summer Triangle owns the zenith – the classic camping sky.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M8 · Lagoon, crossing due south☐**M20 · Trifid, same field**☐**M27 · Dumbbell, overhead**☐**M57 · Ring Nebula, near the zenith**☐**Albireo · high in the Swan**☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[« July](#)[September »](#)

September

The Milky Way tips west while Andromeda climbs the northeast – autumn arrives after midnight first.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M31 · Andromeda, rising northeast☐**M27 · Dumbbell, still overhead**☐**Albireo · high in the west**☐**M13 · Hercules Cluster, west, still fine**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[« August](#)[October »](#)

October

Galaxy of the month: our neighbor, near the zenith. The Orionid meteors peak late in the month.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M31 · Andromeda, near overhead☐**The Double Cluster, high northeast**☐**M45 · Pleiades, up by ten**☐**M27 · Dumbbell, early evening west**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[<< September](#)[November >>](#)

November

Dark by six – the autumn sky at its best with winter rising behind it. Leonid meteors mid-month.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M31 · Andromeda, overhead☐**The Double Cluster, near the zenith**☐**M45 · Pleiades, high east**☐**M42 · Orion Nebula, rises late evening**☐☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[<< October](#)[December >>](#)[<< The hub](#)[THE CATALOG >>](#)

December

The Geminid meteors mid-month are the year's best, and winter's showpieces return in force.

HIGHLIGHTS · TAP FOR THE TARGET CARD

TARGET

EYEPIECE · DATE

SEEN

M42 · Orion Nebula, up by nine☐**M45 · Pleiades, near the zenith**☐**The Double Cluster, overhead**☐**M31 · Andromeda, high west**☐**M44 · Beehive, rising late evening**☐

Unused rows are yours – pencil in doubles, comets, and whatever the night volunteers.

MOON & PLANETS

The Moon and the planets keep their own calendar – check a current almanac for what rides the ecliptic tonight. Saturn, Jupiter, Venus, Mars, and the Moon each hold a card in the catalog, whatever the month.

MOON & PLANET NOTES, THIS YEAR

[<< November](#)[<< The hub](#)[THE CATALOG >>](#)

M31

Andromeda Galaxy

Life list »

Autumn evenings, high overhead

TYPE	Spiral galaxy
CONSTELLATION	Andromeda
MAGNITUDE	3.4 – naked-eye from a dark site
DIFFICULTY	Easy – plain in binoculars

FINDER NOTES

Find the Great Square of Pegasus, then follow the chain of Andromeda two stars east to Mirach. Turn a right angle north through fainter Mu Andromedae and go the same distance again – the long oval glow is the galaxy's core. Binoculars show it even over towns; a telescope at low power adds the bright nucleus and, from dark sites, a dust lane and the companions M32 and M110. The light landing in your eye left it about 2.5 million years ago.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M42

Orion Nebula

The winter showpiece, due south mid-evening

[Life list »](#)

TYPE	Emission nebula
CONSTELLATION	Orion
MAGNITUDE	4.0 – visible to the naked eye
DIFFICULTY	Easy – rewards every instrument

FINDER NOTES

Below Orion's three-star Belt hangs the Sword. Its middle "star" looks fuzzy to the naked eye – that fuzz is the nebula. Any telescope shows gray-green wings of gas folded around the Trapezium, a tight quadrilateral of newborn stars at the heart. Start at low power for the full sweep, then magnify the Trapezium; averted vision doubles how far the wings reach.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M45

Pleiades

Winter evenings, riding high ahead of Orion

[Life list »](#)

TYPE	Open cluster
CONSTELLATION	Taurus
MAGNITUDE	1.6 – an unmistakable naked-eye knot
DIFFICULTY	Easy – binoculars frame it best

FINDER NOTES

Follow Orion's Belt up past orange Aldebaran and keep going; the tight dipper-shaped knot of blue-white stars is the Pleiades, the Seven Sisters. Six or seven show to the naked eye; binoculars spill several dozen across the field, which is the finest way to take them – most telescopes magnify too much to hold the whole cluster at once.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M13

Hercules Cluster

Life list »

Summer evenings, near the zenith

TYPE	Globular cluster
CONSTELLATION	Hercules
MAGNITUDE	5.8 – a gray puff in a finder scope
DIFFICULTY	Easy – resolves with aperture

FINDER NOTES

Find the Keystone, the four-star trapezoid of Hercules, high overhead on summer evenings. M13 sits on its western side, about a third of the way down from Eta toward Zeta Herculis – binoculars show a fuzzy star. A 150 mm telescope at around 100x begins to resolve its edges into pinpoint stars: a swarm of several hundred thousand suns, about 25,000 light-years out.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M8

Lagoon Nebula

Life list »

Summer, low in the south over the Teapot

TYPE	Emission nebula
CONSTELLATION	Sagittarius
MAGNITUDE	6.0 – naked-eye under dark skies
DIFFICULTY	Moderate – wants a clear south horizon

FINDER NOTES

Above the spout of the Sagittarius Teapot, dark-sky observers see a small brightening in the Milky Way with the naked eye. Binoculars show a glowing pool cut by a dark channel – the lagoon – with the young cluster NGC 6530 embedded on its eastern side. From mid-northern latitudes it never climbs high, so catch it in the hour it crosses due south.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M20

Trifid Nebula

[Life list »](#)

Summer, in the same binocular field as the Lagoon

TYPE	Emission and reflection nebula
CONSTELLATION	Sagittarius
MAGNITUDE	6.3 – small and low
DIFFICULTY	Challenging – dark skies help

FINDER NOTES

Sweep about a degree and a half north-northwest from the Lagoon Nebula; the two share a low-power field. A telescope from dark sites shows a round glow split by dark lanes into three lobes – the "trifid" of the name – with a fainter blue reflection cap on its north side. Haze or moonlight erases it, so save this one for a transparent, moonless night.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M27

Dumbbell Nebula

[Life list »](#)

Summer and early autumn, high in Vulpecula

TYPE	Planetary nebula
CONSTELLATION	Vulpecula
MAGNITUDE	7.4 – bright for its class
DIFFICULTY	Easy in a small telescope

FINDER NOTES

From Gamma Sagittae, the tip star of the little Arrow, nudge about three degrees north. At low power a bright apple-core of glowing gas appears – the brightest planetary nebula in the northern sky, the cast-off shroud of a dying sunlike star. It takes magnification well; an OIII or UHC filter makes the two bright lobes stand out sharply.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M57

Ring Nebula

[Life list »](#)

Summer, between the bottom stars of Lyra

TYPE	Planetary nebula
CONSTELLATION	Lyra
MAGNITUDE	8.8 – small but high-contrast
DIFFICULTY	Moderate – needs about 100x

FINDER NOTES

Point midway between Beta and Gamma Lyrae, the bottom pair of Lyra's parallelogram hanging under Vega. The Ring is too small for binoculars – at low power it is one "star" among many – but at 100x it becomes a tiny, perfect gray smoke ring. A classic test of steady seeing: the sharper the night, the crisper the hole in the middle.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M81

Bode's Galaxy

Spring evenings, high in the north

Life list »

TYPE	Spiral galaxy
CONSTELLATION	Ursa Major
MAGNITUDE	6.9 – binocular-bright from dark sites
DIFFICULTY	Moderate – an easy star-hop

FINDER NOTES

Draw a diagonal across the Big Dipper's bowl from Phecda through Dubhe and extend it the same distance again. M81 waits there as an oval glow – and the cigar-shaped starburst galaxy M82 shares the same low-power field, two galaxies in one eyepiece. The pair is circumpolar from mid-northern latitudes, best when the Dipper rides highest on spring evenings.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M44

Beehive Cluster

Life list »

Late winter into spring, midway up the south

TYPE	Open cluster
CONSTELLATION	Cancer
MAGNITUDE	3.7 – a naked-eye mist
DIFFICULTY	Easy – made for binoculars

FINDER NOTES

Halfway along the line from Pollux in Gemini to Regulus in Leo, inside the dim trapezoid of Cancer. The naked eye sees a misty patch – the ancients called it Praesepe, the Manger – and binoculars burst it into dozens of stars. Like the Pleiades it is wide: use the lowest power you own, or the hive flies out of the field.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M51

Whirlpool Galaxy

[Life list »](#)

Spring evenings, under the Dipper's handle

TYPE	Interacting spiral galaxy
CONSTELLATION	Canes Venatici
MAGNITUDE	8.4 – surface brightness is low
DIFFICULTY	Challenging – dark skies required

FINDER NOTES

From Alkaid, the end star of the Big Dipper's handle, drop about three and a half degrees toward Cor Caroli. Under dark skies a telescope shows two touching glows: the face-on spiral and its small companion NGC 5195, caught mid-collision. Medium apertures on moonless nights begin to show the spiral arms – the first galaxy ever seen to have them.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



M104

Sombrero Galaxy

[Life list »](#)

Spring evenings, midway up the southern sky

TYPE	Edge-on spiral galaxy
CONSTELLATION	Virgo
MAGNITUDE	8.0 – condensed and starlike at first
DIFFICULTY	Challenging – small and southerly

FINDER NOTES

About eleven degrees west of Spica, on the border between Virgo and Corvus. A telescope shows a bright spindle with a bulging core; from dark sites, larger apertures reveal the dark dust lane slicing along its edge – the brim of the hat. It sits well south, so take it as it crosses the meridian, and let the eye settle before judging what you see.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



NGC 869/884

The Double Cluster

[Life list »](#)

Autumn and winter, high between Perseus and Cassiopeia

TYPE	Pair of open clusters
CONSTELLATION	Perseus
MAGNITUDE	4.3 together – naked-eye from dark skies
DIFFICULTY	Easy – a low-power showpiece

FINDER NOTES

Halfway between the W of Cassiopeia and the bright stretch of Perseus, the naked eye picks up an elongated soft glow in the Milky Way. Binoculars split it into two rich clusters sharing one field; a telescope at the lowest power drips with stars, including a scatter of orange supergiants. Circumpolar from mid-northern latitudes and at its best on autumn and winter evenings.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



Beta Cyg

Albireo

Summer, at the head of the Swan

Life list »

TYPE	Double star – gold and sapphire
CONSTELLATION	Cygnus
MAGNITUDE	3.1 combined – components 3.4 and 5.1
DIFFICULTY	Easy – splits at low power

FINDER NOTES

The star at the head of Cygnus the Swan – the foot of the Northern Cross – where the constellation points down the summer Milky Way. Any telescope at about 30x splits it into a golden primary with a blue-white companion, the sky's finest color-contrast pair. If the two stars will not separate, nudge the power up a step and let the seeing settle.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



PLANET

Saturn

[Life list »](#)

Wanders the zodiac – check an almanac for tonight

TYPE	Ringed gas giant
CONSTELLATION	Moves along the ecliptic
MAGNITUDE	About -0.5 to 1.0 – ring tilt decides
DIFFICULTY	Easy – rings show from 25x

FINDER NOTES

A steady, cream-colored point that does not twinkle like the stars around it; a current almanac, planisphere, or app gives its place among the zodiac constellations tonight. Any telescope at 25x shows the rings – the sight that has hooked observers for four centuries. At 100x in steady air look for the Cassini Division, the shadow of the globe on the rings, and 8th-magnitude Titan standing off to one side. The ring tilt opens and closes over a 29-year cycle, so no two apparitions look quite alike.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



PLANET

Jupiter

[Life list »](#)

Wanders the zodiac – check an almanac for tonight

TYPE	Gas giant with four bright moons
CONSTELLATION	Moves along the ecliptic
MAGNITUDE	About -2.0 to -2.9 – outshines every star
DIFFICULTY	Easy – moons even in binoculars

FINDER NOTES

When it is up, usually the brightest starlike object in the late-evening sky – only Venus outshines it. Steadied binoculars already show up to four Galilean moons in a line that rearranges night to night; a telescope at 100x shows the two dark equatorial cloud belts, and in good seeing the Great Red Spot as the planet's ten-hour rotation carries it across. Watch for moon shadows crossing the disk – a current almanac lists the events.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



PLANET

Venus

[Life list »](#)

The evening or morning star – check an almanac

TYPE	Inner planet – shows phases
CONSTELLATION	Never far from the Sun
MAGNITUDE	About -4 – the brightest point of light
DIFFICULTY	Easy – but mind the nearby Sun

FINDER NOTES

The blazing "evening star" low in the west after sunset, or the "morning star" before dawn – Venus never strays more than about 47 degrees from the Sun, so it is always a twilight object. A telescope shows a brilliant white phase like a miniature Moon: gibbous when far around its orbit, a large thin crescent when it swings near. Wait until the Sun is fully below the horizon before pointing any optics its way.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



PLANET

Mars

[Life list »](#)

Wanders the zodiac – best near opposition

TYPE	Rocky planet – the red wanderer
CONSTELLATION	Moves along the ecliptic
MAGNITUDE	From -2.9 at a close opposition to 1.8
DIFFICULTY	Moderate – detail wants opposition

FINDER NOTES

An unmistakable rust-orange point when it is up; an almanac gives tonight's position. Mars only shows a worthwhile disk in the months around its oppositions, which come roughly every 26 months – between them it shrinks to a featureless dot. Near opposition, 150x to 250x in steady air can show a white polar cap and dusky surface markings; be patient and observe on several nights, since the planet rotates only a little slower than Earth.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



LUNA

The Moon

[Life list »](#)

Best along the terminator – first quarter especially

TYPE	Natural satellite
CONSTELLATION	Crosses the zodiac each month
MAGNITUDE	About -12.7 when full
DIFFICULTY	Easy – every instrument rewards

FINDER NOTES

No finding required – the craft is timing. Look along the terminator, the moving line between lunar day and night, where low sunlight throws every crater and ridge into sharp relief. First quarter, high in the evening sky, is the classic session; full Moon is flat and blinding by comparison. Work the terminator north to south at rising powers, and use a Moon filter if the glare stings. The features are endless: no two nights repeat.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



SAFETY

The Sun in Eclipse

[Life list »](#)

Read before any solar attempt – no exceptions

TYPE	Solar eclipse – safety brief
CONSTELLATION	Wherever the Moon crosses the Sun
MAGNITUDE	Never observe it unfiltered
DIFFICULTY	Certified solar filters only

FINDER NOTES

Never look at the Sun – eclipsed or not – without proper protection, and never through any telescope, binocular, or camera unless a certified solar filter is fitted over the FRONT of the instrument. For your eyes alone, use eclipse glasses that meet ISO 12312-2, undamaged and from a reputable maker. Sunglasses, smoked glass, exposed film, and eyepiece-end "sun filters" are never safe. The safest method needs no filter at all: pinhole projection, casting the Sun's image through a small hole onto paper. Only during the brief totality of a total eclipse, with the Sun's disk completely covered, may filters come off – and they go back on before the first sliver of Sun returns.

FIRST OBSERVED

DATE

INSTRUMENT · EYEPIECE

SKY · SEEING

NOTES



The Life List

Twenty sights, ticked off as the seasons hand them over – date and instrument beside each.

TARGET · TAP FOR ITS CARD	FIRST SEEN · WITH
☐ M31 · Andromeda Galaxy	
☐ M42 · Orion Nebula	
☐ M45 · Pleiades	
☐ M13 · Hercules Cluster	
☐ M8 · Lagoon Nebula	
☐ M20 · Trifid Nebula	
☐ M27 · Dumbbell Nebula	
☐ M57 · Ring Nebula	
☐ M81 · Bode's Galaxy	
☐ M44 · Beehive Cluster	
☐ M51 · Whirlpool Galaxy	
☐ M104 · Sombrero Galaxy	
☐ NGC 869/884 · The Double Cluster	
☐ Beta Cyg · Albireo	
☐ Saturn	
☐ Jupiter	
☐ Venus	
☐ Mars	
☐ The Moon	
☐ The Sun in Eclipse · safety	



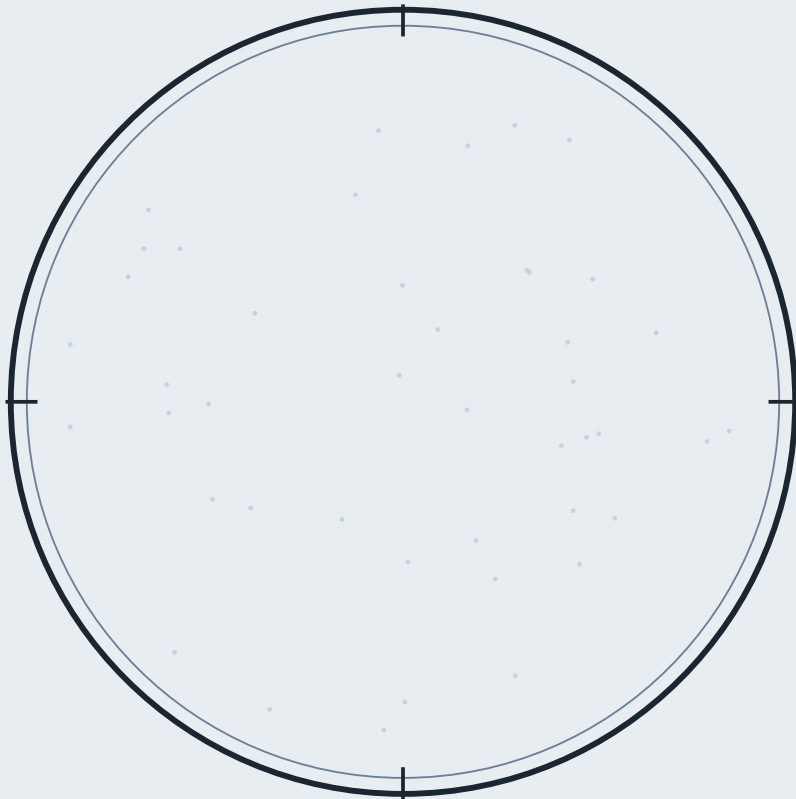
Session 01

DATE

SKY · CONDITIONS

SEEING

TARGET

[The catalog »](#)

SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[Session 02 »](#)[« The hub](#)[LIFE LIST »](#)

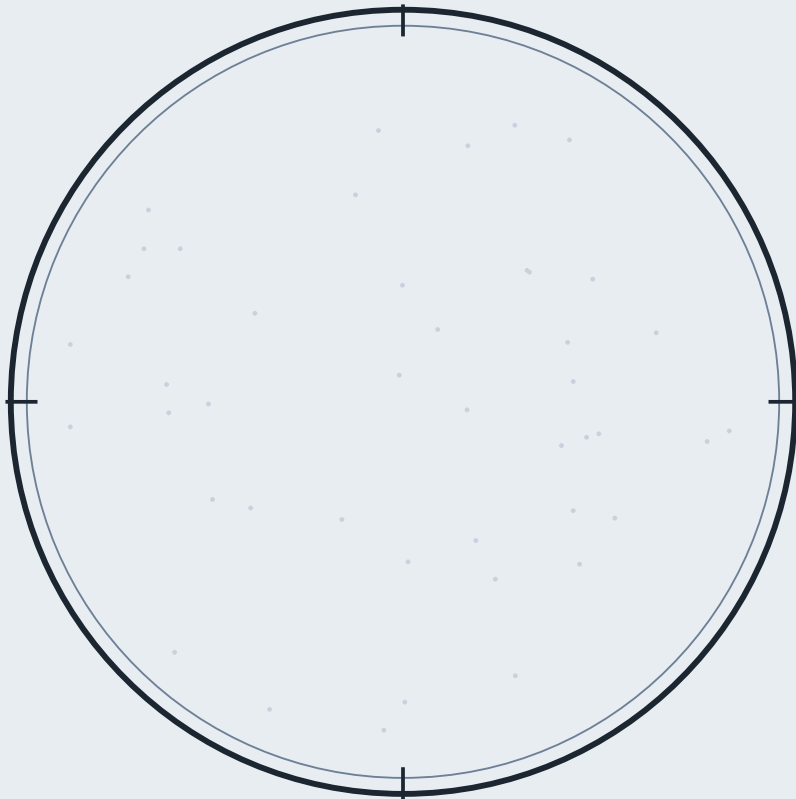
Session 02

DATE

SKY · CONDITIONS

SEEING

TARGET

[The catalog »](#)

SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 01](#)[Session 03 »](#)[« The hub](#)[LIFE LIST »](#)

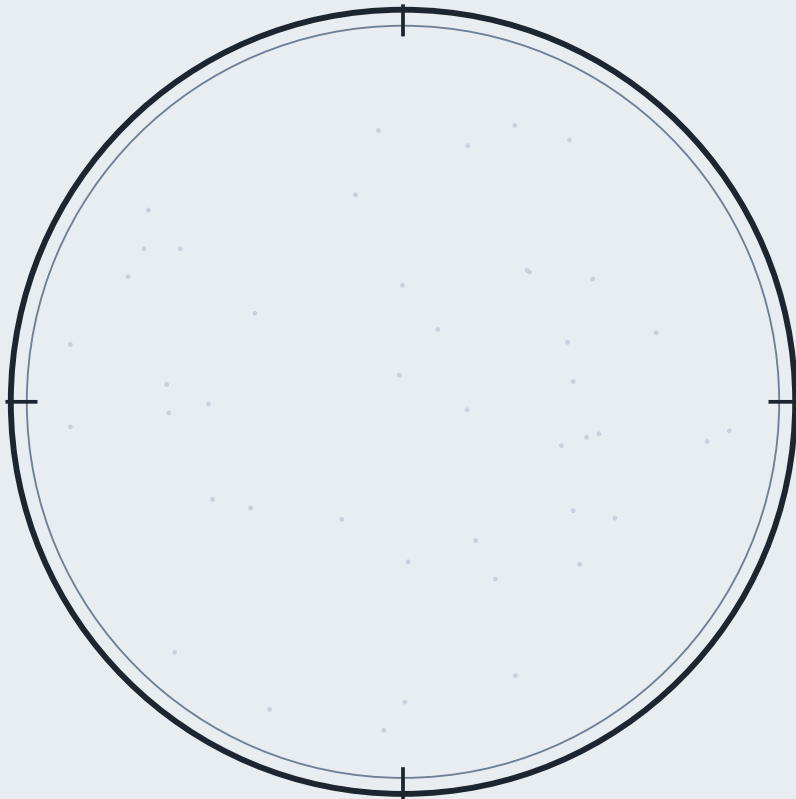
Session 03

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 02](#)[Session 04 »](#)[« The hub](#)[LIFE LIST »](#)

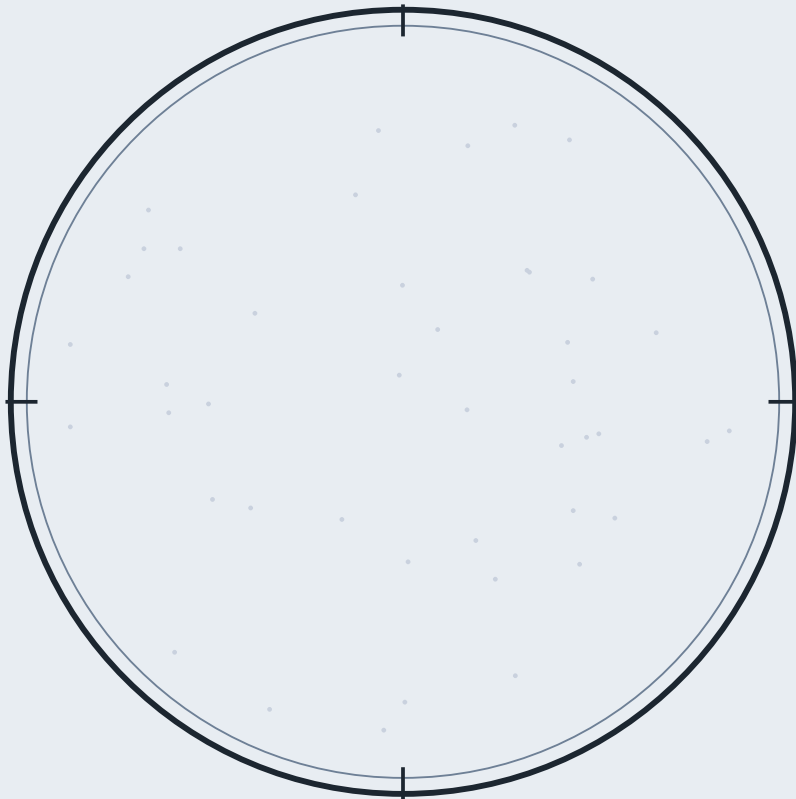
Session 04

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 03](#)[Session 05 »](#)[« The hub](#)[LIFE LIST »](#)

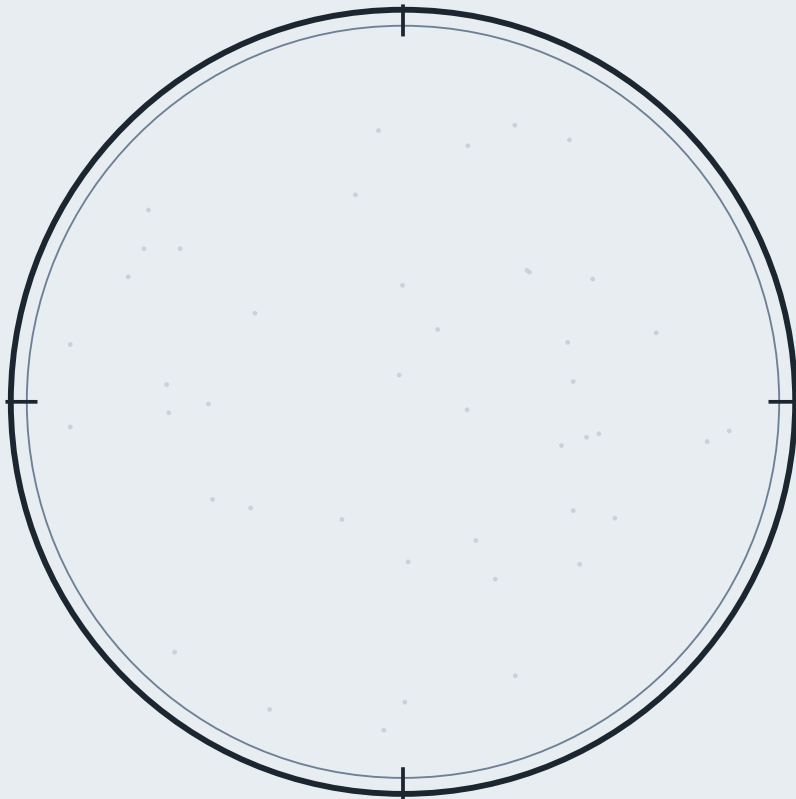
Session 05

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 04](#)[Session 06 »](#)[« The hub](#)[LIFE LIST »](#)

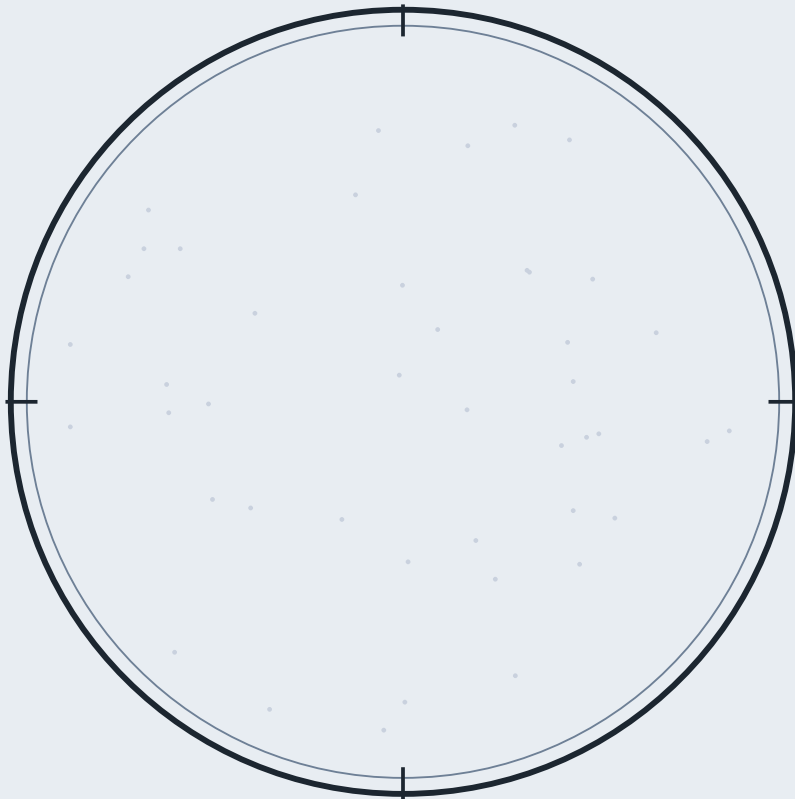
Session 06

DATE

SKY · CONDITIONS

SEEING

TARGET

[The catalog »](#)

SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 05](#)[Session 07 »](#)[« The hub](#)[LIFE LIST »](#)

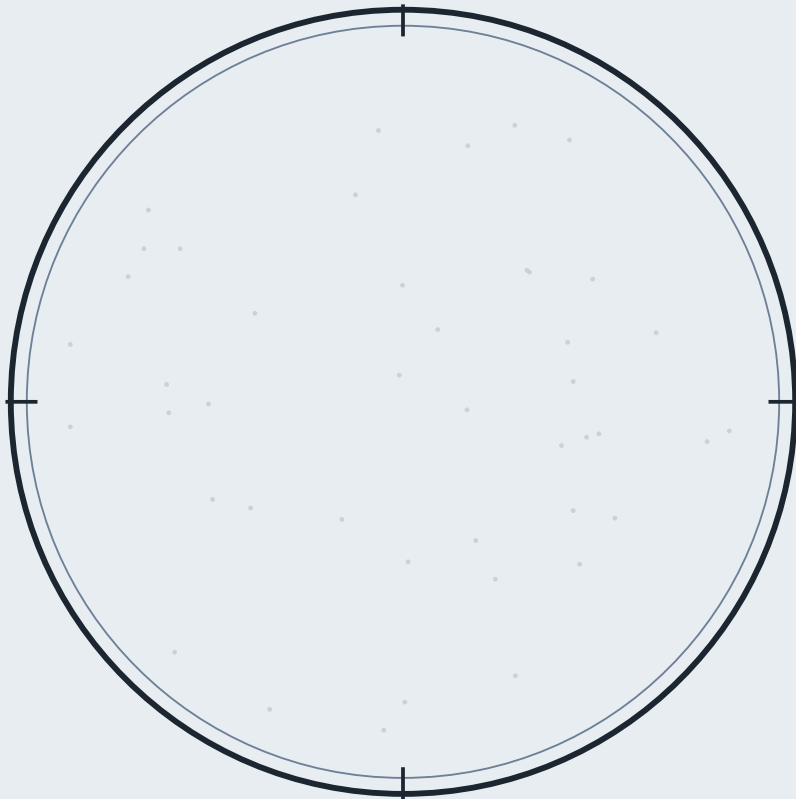
Session 07

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 06](#)[Session 08 »](#)[« The hub](#)[LIFE LIST »](#)

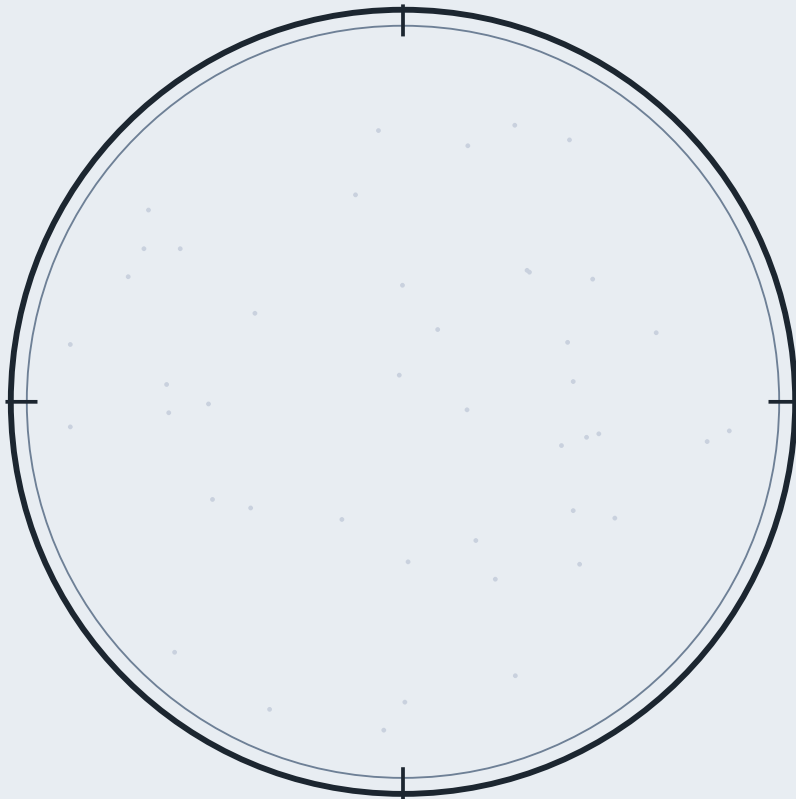
Session 08

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 07](#)[Session 09 »](#)[« The hub](#)[LIFE LIST »](#)

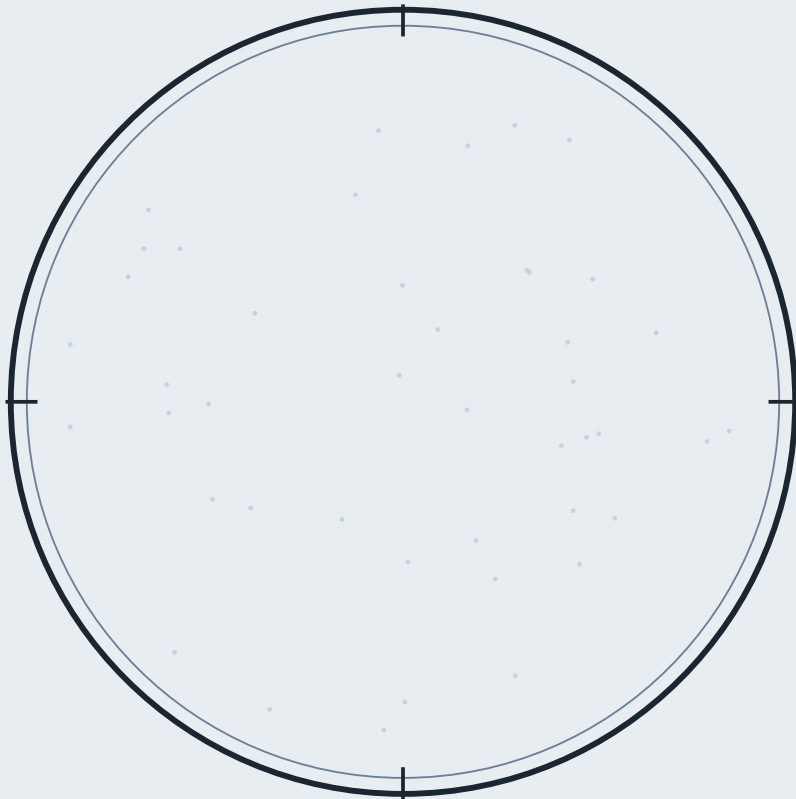
Session 09

DATE

SKY · CONDITIONS

SEEING

TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 08](#)[Session 10 »](#)[« The hub](#)[LIFE LIST »](#)

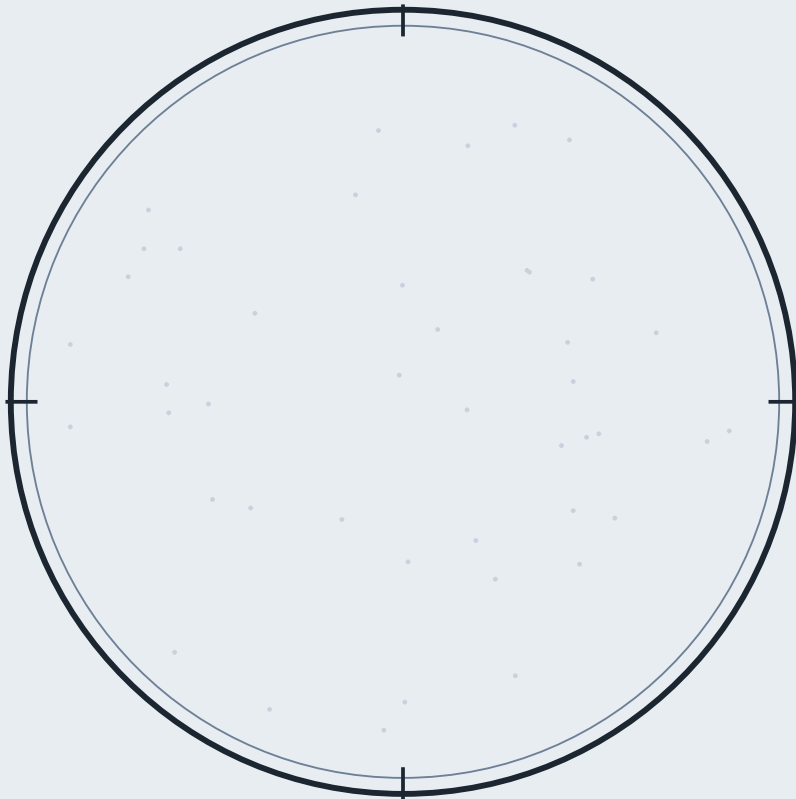
Session 10

DATE

SKY · CONDITIONS

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 09](#)[Session 11 »](#)[« The hub](#)[LIFE LIST »](#)

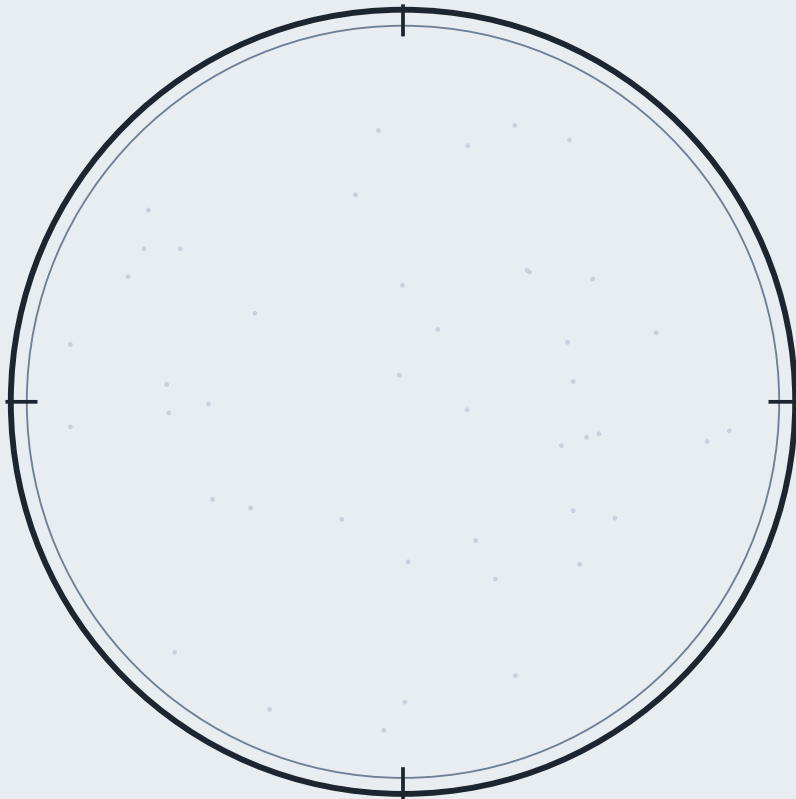
Session 11

DATE

SKY · CONDITIONS

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

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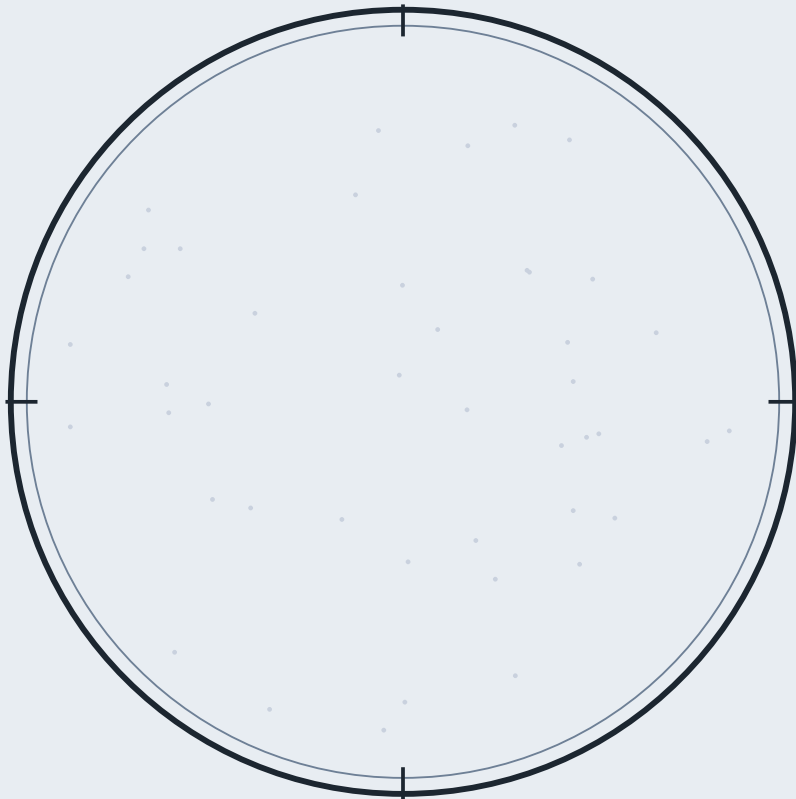
Session 12

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SKY · CONDITIONS

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TARGET

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

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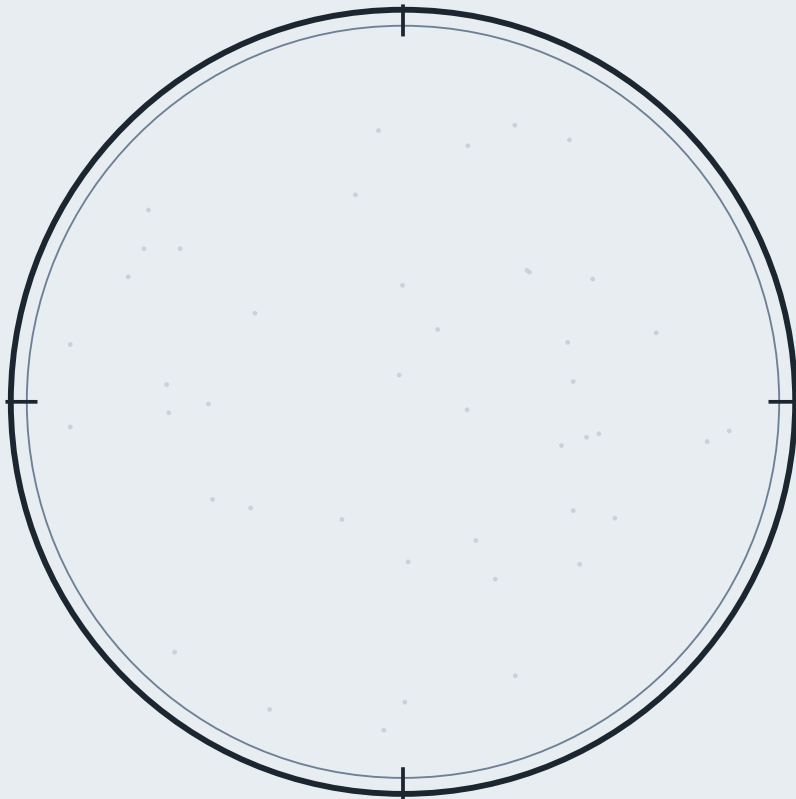
Session 13

DATE

SKY · CONDITIONS

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 12](#)[Session 14 »](#)[« The hub](#)[LIFE LIST »](#)

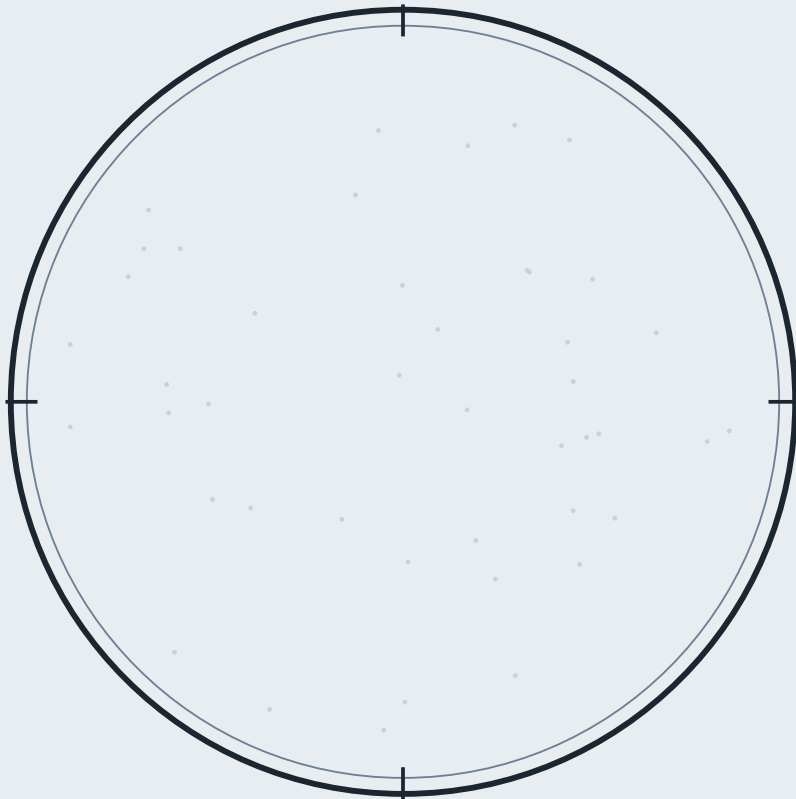
Session 14

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 13](#)[Session 15 »](#)[« The hub](#)[LIFE LIST »](#)

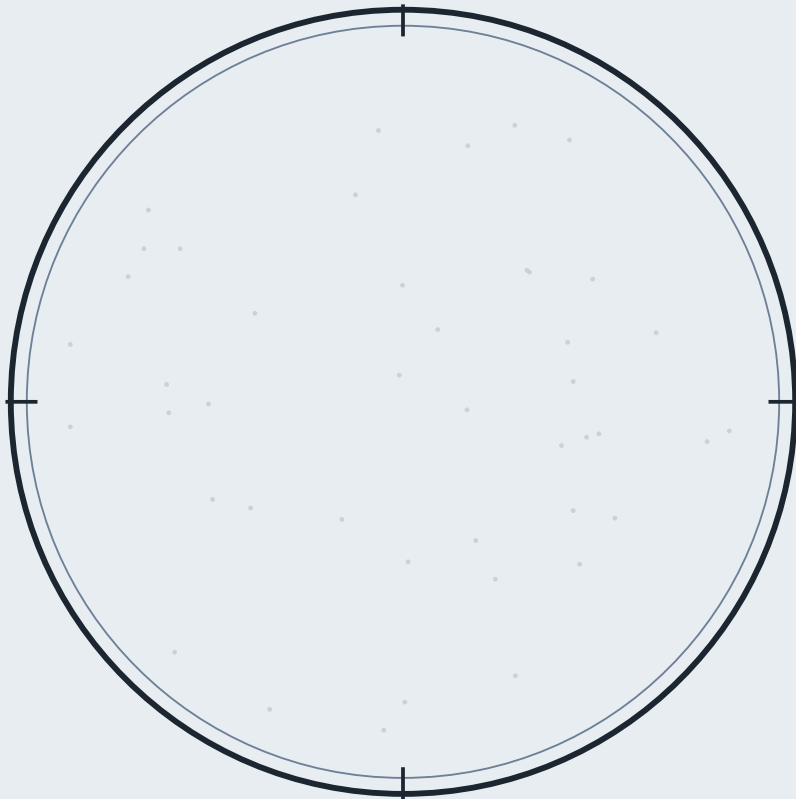
Session 15

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 14](#)[Session 16 »](#)[« The hub](#)[LIFE LIST »](#)

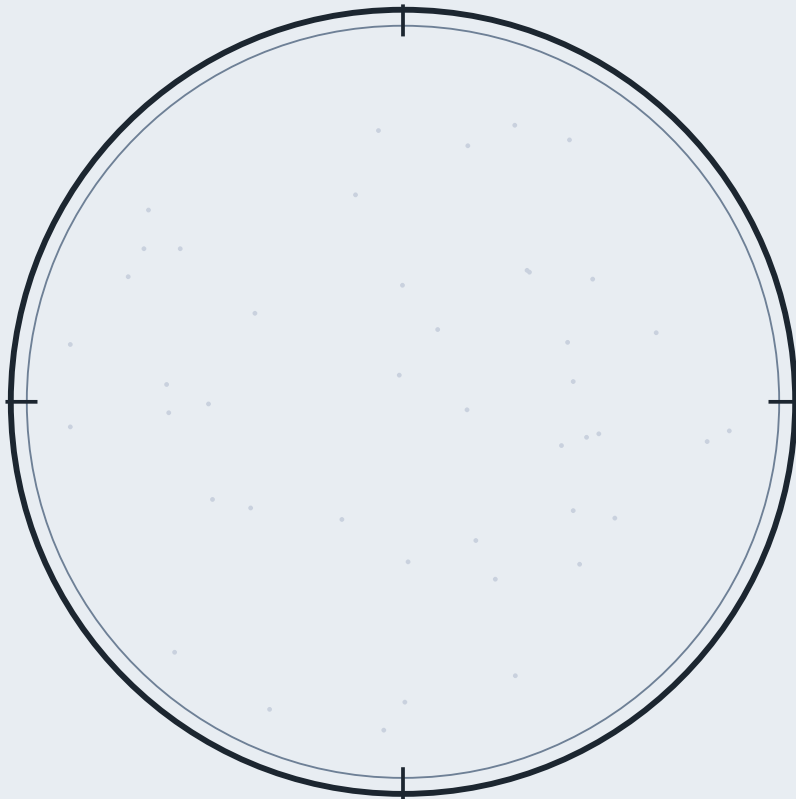
Session 16

DATE

SKY · CONDITIONS

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 15](#)[Session 17 »](#)[« The hub](#)[LIFE LIST »](#)

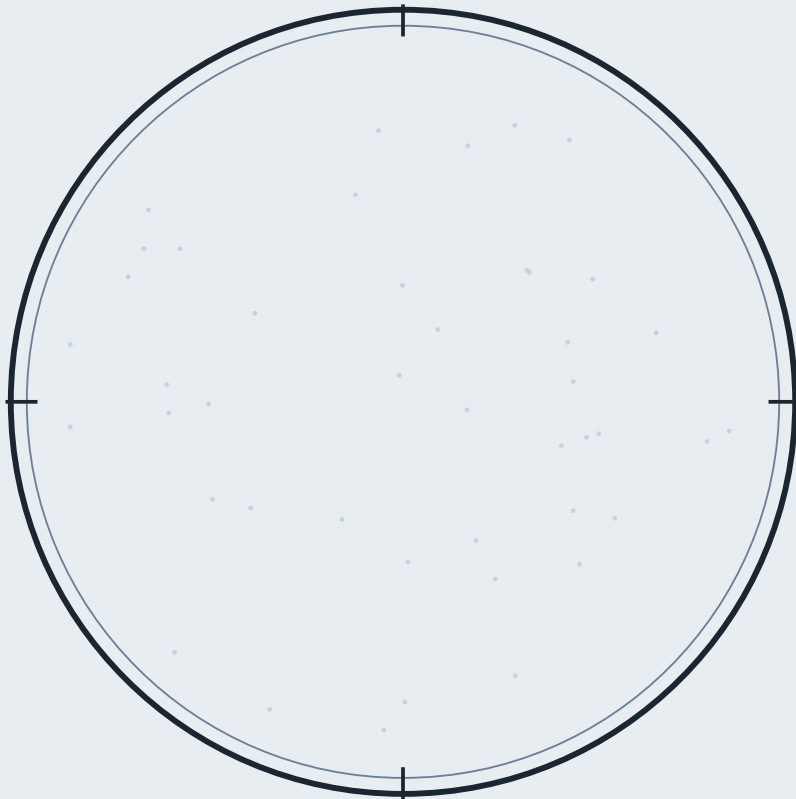
Session 17

DATE

SKY · CONDITIONS

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 16](#)[Session 18 »](#)[« The hub](#)[LIFE LIST »](#)

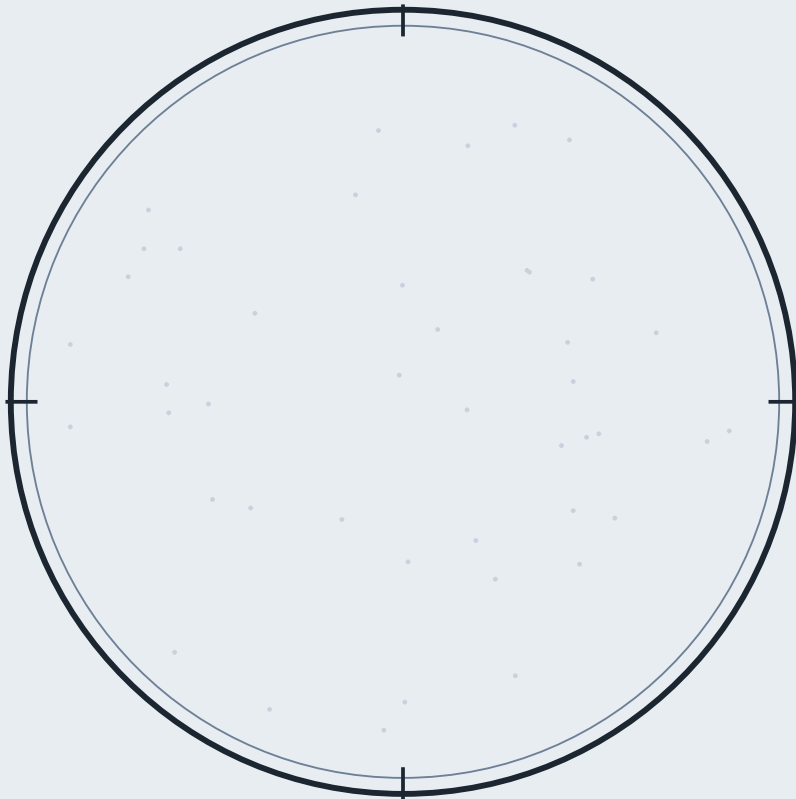
Session 18

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 17](#)[Session 19 »](#)[« The hub](#)[LIFE LIST »](#)

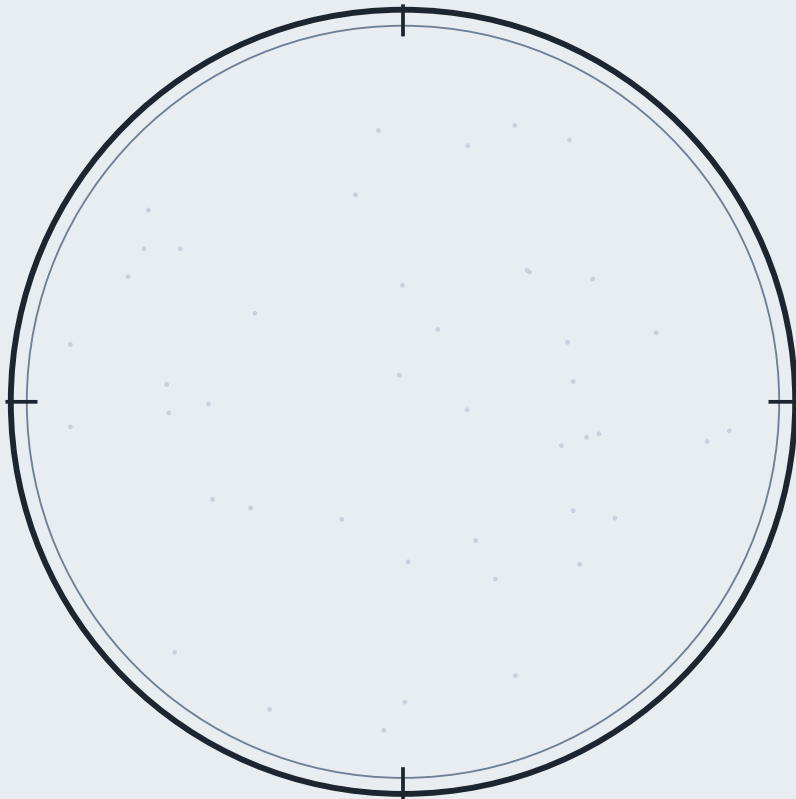
Session 19

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

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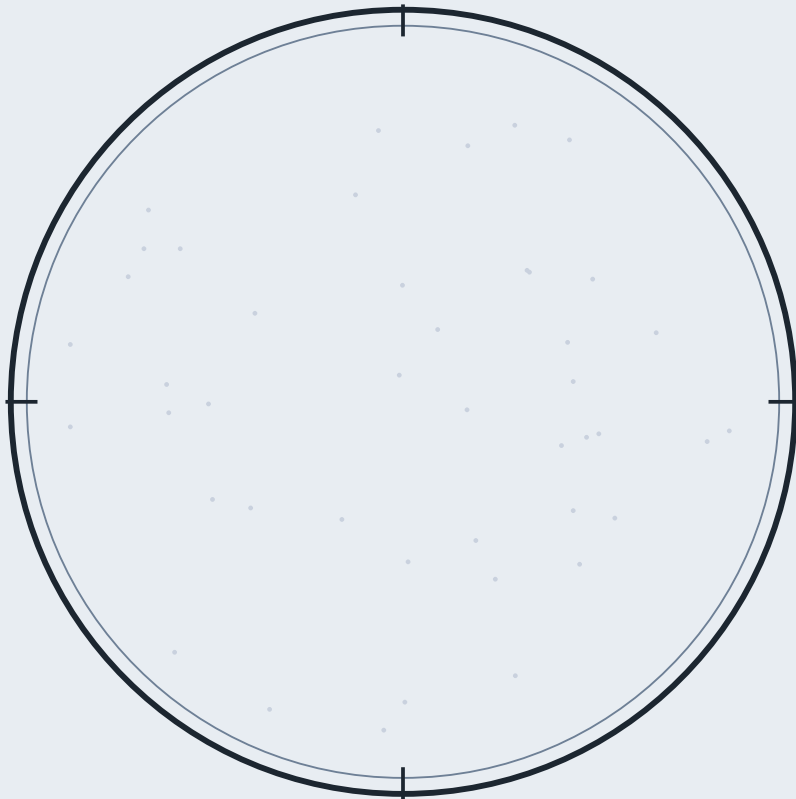
Session 20

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SKETCH THE FIELD · MARK NORTH ON THE RIM YOURSELF

THE NIGHT, IN WORDS

[« Session 19](#)[« The hub](#)[LIFE LIST »](#)

The Equipment Shelf

The kit, written down once – so the log can say "the 10 mm" and mean it.

OPTICS

INSTRUMENT	APERTURE	FOCAL LENGTH	NOTES

EYEPIECES & FILTERS

EYEPIECE	FL (MM)	POWER	NOTES

Power = telescope focal length / eyepiece focal length. Exit pupil = aperture / power – keep it between about 1 and 7 mm.

THE FIELD KIT

☐	_____	☐	_____
☐	_____	☐	_____
☐	_____	☐	_____

Glossary · A to L

The words the log keeps using, pinned down.

Averted vision

Looking slightly to one side of a faint object. The eye's off-center rods see deeper than its middle, so the nebula brightens when you stop staring at it.

Aperture

The diameter of a telescope's main lens or mirror. It sets how much light you gather – the one number that matters most.

Collimation

The alignment of a telescope's optics. A reflector only delivers sharp stars when collimated; check it when high-power views turn soft.

Dark adaptation

The eyes' slow gain in night sensitivity – near-full after about thirty minutes. One glance at a white light resets the clock; dim red light preserves it.

Declination

The sky's north-south coordinate, latitude's counterpart, in degrees from the celestial equator. Right ascension is its east-west partner, measured in hours.

Ecliptic

The Sun's yearly track against the stars. The Moon and planets keep close to this line – which is why the wanderers always cross the zodiac.

Exit pupil

The width of the light beam leaving the eyepiece: aperture divided by power. Between about 1 and 7 mm it matches what a dark-adapted eye can use.

Field of view

The patch of sky the eyepiece shows. Big showpieces – the Pleiades, the Beehive, the Double Cluster – want the widest, lowest-power field you own.

Globular cluster

A dense ball of hundreds of thousands of ancient stars orbiting the galaxy's core. M13 in Hercules is the northern sky's showpiece.

M to Z »



Glossary · M to Z

The words the log keeps using, pinned down.

Magnitude	Brightness on an inverted scale – bigger numbers are fainter. Five magnitudes is a hundredfold step; magnitude 6 is the naked-eye limit under dark skies.
Messier catalog	Charles Messier's 18th-century list of about 110 comet-impostors – nebulae, clusters, and galaxies that are now the beginner's grand tour of the deep sky.
Open cluster	A loose family of young stars born from one cloud and still traveling together. The Pleiades wrote the type specimen.
Opposition	When an outer planet stands opposite the Sun in our sky – rising at sunset, up all night, closest, and brightest. The months to watch Mars.
Planetary nebula	The glowing cast-off shell of a dying sunlike star – nothing to do with planets beyond the round look in old telescopes. The Ring and the Dumbbell are the classics.
Seeing	The steadiness of the air. Good seeing means still, sharp stars at high power; poor seeing smears them. Judge it every session – it decides your magnification.
Terminator	The line between day and night on the Moon, where shadows run longest and craters stand in sharpest relief. Observe along it, not at the full glare.
Transparency	The clarity of the air – how faint you can see. Haze and high moisture kill transparency on nights that look cloudless; the faint galaxies notice first.
Zenith	The point straight overhead, where you look through the least air. Targets near it show the sky at its darkest and steadiest.

« A to L

« The hub



EQUIPMENT »